Applied Surface Science 640 (2023) 158327

Contents lists available at ScienceDirect

Applied Surface Science

journal homepage: www.elsevier.com/locate/apsusc

Full Length Article

Mechanism and optimization of femtosecond laser welding fused silica and aluminum

Jie Zhan [a], Yuhang Gao [a], Jiaxin Sun [a], Weihua Zhu [a], Sumei Wang [a,b,*], Lan Jiang [a,b,c], Xin Li [a]

[a] Laser Micro/Nano Fabrication Laboratory, School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China
[b] Yangtze Delta Region Academy of Beijing Institute of Technology, Jiaxing 314000, China
[c] Beijing Institute of Technology Chongqing Innovation Center, Chongqing 401120, China

ARTICLE INFO

Keywords:
Femtosecond laser welding
Optical contact
Self-focusing
Pre-compensation

ABSTRACT

The ability to weld metals and insulators is widely required in aerospace, sensors and Micro-electro-mechanical systems (MEMS). Femtosecond laser welding is a precise and clean technique for bonding materials with benefits including high location accuracy, low heat-affected zone and little waste generation. However, the connection process is still unclear so that pump–probe microscopy and plasma image were used to reveal the welding mechanism at the contact interface. Under the irradiation of a femtosecond laser pulse, the aluminum melted after a few picoseconds, then re-solidified and connected two different materials. Additionally, plasma was limited in the interaction zone after laser irradiation for optical contact (OC), so the smaller affected region with the most absorbed energy was jointed easily with higher shear strength. Self-focusing and non-linear absorption in fused silica are key factors affecting the shear strength, since the energy absorption and modification in fused silica would induce internal stress and decrease the energy deposition at the contact interface. Our findings not only reduce the fracture of fused silica and increase shear strength up to 25.75 MPa by means of focal point pre-compensation, but also demonstrate the extraordinary potential of ultrashort laser welding in the field of precise connection of heterogenous materials.

1. Introduction

The ability to connect heterogeneous materials is significantly important for electronic components in systems such as precision sensors [1,2], lab-on-a-chip [3], and Micro-electro-mechanical systems (MEMS) [4]. However, traditional bonding methods (direct bonding [5], anodic bonding [3,6], and adhesive bonding [7]) have large inherent limitations (aging, degassing, and lack of bonding accuracy), which do not match the needs of micro/nano manufacturing. Presently, continuous laser and nanosecond laser, which are commonly used in laser welding, have the advantage of high positioning accuracy. However, they are based on preheating and slow cooling to complete the connection, which would cause obvious thermal deformations on samples, and defects like cracks and pores at weld seams are inevitable [8–12].

The ability of the femtosecond laser to weld heterogeneous, hard, and brittle materials has attracted extensive attention [13]. Additionally, its precise connection position and small heat-affected zone have broad application prospects in precision packaging [14]. Although femtosecond laser welding has been applied to connect heterogeneous materials (such as fused silica and various materials [15–18], silicon and copper [19], and glass with different compositions [20,21]), its connection mechanism has not been seriously studied. The propagation process inside transparent materials changes from linear to nonlinear with the increased laser intensity. The optical Kerr effect and multiphoton ionization mainly affect the nonlinear propagation [22,23]. It is manifested as the forward focal shift (self-focusing) and the generation of multiple focal points, reducing the energy deposition at the weld seam and leading to the internal damage of the transparent material. Therefore, the beam propagation in the medium keeps collapsing inward due to the Gaussian distribution of light intensity, and the location of the focus appears earlier than the geometric focus [19,24]. Subsequently, the plasma produced at the focal point diverges the Gaussian beam. However, the beam may focus again and undergo another focusing-defocusing cycle, causing multiple focal points [22,25].

This study tried to illustrate the mechanism of femtosecond laser welding of heterogeneous materials based on femtosecond laser welding of aluminum and fused silica. Furthermore, it optimized the welding

* Corresponding author at: Laser Micro/Nano Fabrication Laboratory, School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China.
E-mail address: wangsumei@bit.edu.cn (S. Wang).

https://doi.org/10.1016/j.apsusc.2023.158327
Received 6 June 2023; Received in revised form 15 August 2023; Accepted 24 August 2023
Available online 25 August 2023

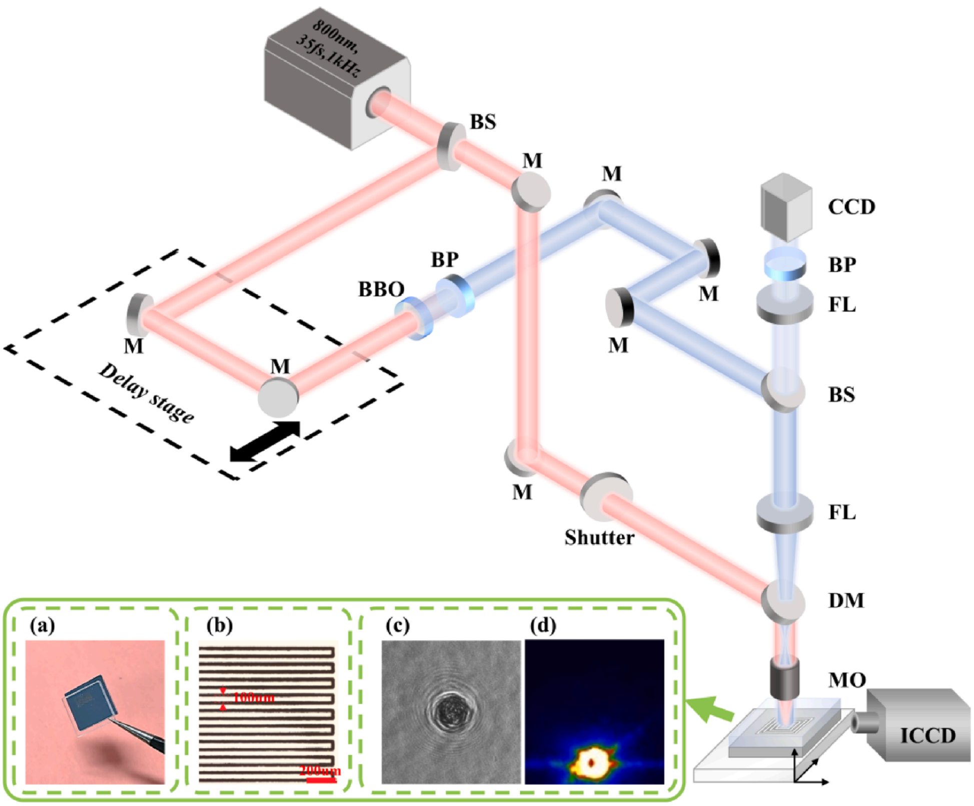

Fig. 1. Experimental scheme for weld and in-situ observation. BS, M, BBO, BP, CCD, FL, DM, MO, and ICCD represent the beam splitter, mirror, beta barium boron oxide crystal, band-pass filter, charge-coupled device, focal lens, dichroic mirror, microscope objective, and intensified charge-coupled device, respectively. The center wavelength and bandwidth of the band-pass filter are 400 nm and 10 nm. The insets show the results of processing and observation: a) The weld sample; b) An optical micrograph at the weld area; c) The pump–probe result of a single pulse under welding conditions; d) The plasma image captured from the polished side of a sample.

results to improve the shear strength of the samples and reduce the fused silica inner damage. The interaction of femtosecond laser at the optical contact (OC) interface of fused silica and aluminum was studied using the pump–probe technique, which is commonly used in recording and analysing the transient evolution for both the time and space domains on materials surfaces [26–33]. The interaction mechanism between laser and bilayer materials at different energies was predicted using the reflectance change. The weld spot presented a ring structure in a steady state, similar to that in liquid ablation, due to non-free expansion at the irradiation spot [31]. The self-focusing phenomenon of femtosecond laser inside fused silica was explained by observing the position and morphology of plasma. Finally, the laser energy deposition at the interface was improved to the maximum through pre-compensation to avoid plasma separation in fused silica. The materials were analyzed by combining shear stress and microscopic morphology to explain related physical mechanisms.

2. Material and methods

2.1. Sample preparation

The materials used in femtosecond laser welding were single-sided polished 10 mm × 10 mm × 1 mm pure commercial aluminum (99.7% Al), double-sided polished 10 mm × 10 mm × 1 mm fused silica (for shear stress text and pump–probe observation), and four-sided polished 5 mm × 5 mm × 1 mm fused silica (for plasma image capture). The surface roughness (Sa) is better than 10 nm with the polished mirror finish surfaces of aluminum and fused silica. All the polished samples are supplied by Hefei Branch Crystal Material Technology Co. Ltd. The laser passed through the upper layer of fused silica and placed the focus spot at the contact interface, between the fused silica and the aluminum samples, with energy under critical power for self-focusing to keep geometric focus at the interface. During welding, the OC was maintained between fused silica and aluminum samples. The OC means the distance between two contact surfaces is considerable small than the wavelength (generally considered to be $\leq \lambda/4$) [34–36]. The size of the welding area was 4 mm × 2 mm, and the row spacing was 100 μm.

2.2. Experimental setup

The experimental scheme is displayed in Fig. 1. A femtosecond laser beam with 800 nm wavelength and 35 fs pulse duration was divided into two using a beam splitter for the pump and probe pulses. The beam diameter at the entrance pupil was limited to 6 mm. A 5 × objective lens (numerical aperture, NA = 0.15, Olympus MPlanFLN 5 ×) and a precise CCD (Imagesource 23U445) were used for welding and observation. The single pump pulse acting on the contact interface of the samples, combined with the probe pulse, could be used to detect the reflectivity evolution of the welding spot. With multiple pulses, an intensified charge-coupled device (ICCD, iStar 334, Andor) was used to collect plasma luminescence signals from the polished side of the sample with 5 ns gate width. Continuous welding experiments with the repetition rate of 1 kHz could also be completed with this system.

2.3. Shear test and bonding area characterization

An electronic universal testing machine was used to test the shear stress of the samples. Additionally, the clamp was designed to hold the samples vertically to keep the pressure direction parallel to the interface. The speed of applying compressive force in the experiment was controlled to 0.2 mm/min to avoid any impact on the samples. The shear stress value was simply defined as the radio between compressive force at which the samples are separated and the welding area (4 × 2 mm^2). More details are shown in Fig. S1 in the Supporting Information. After the shear test, the disassembled samples were characterized using scanning electron microscopy (SEM) and energy dispersive X-ray spectroscopy (EDS) to analyze the microstructures of the weld seams and the elemental compositions.

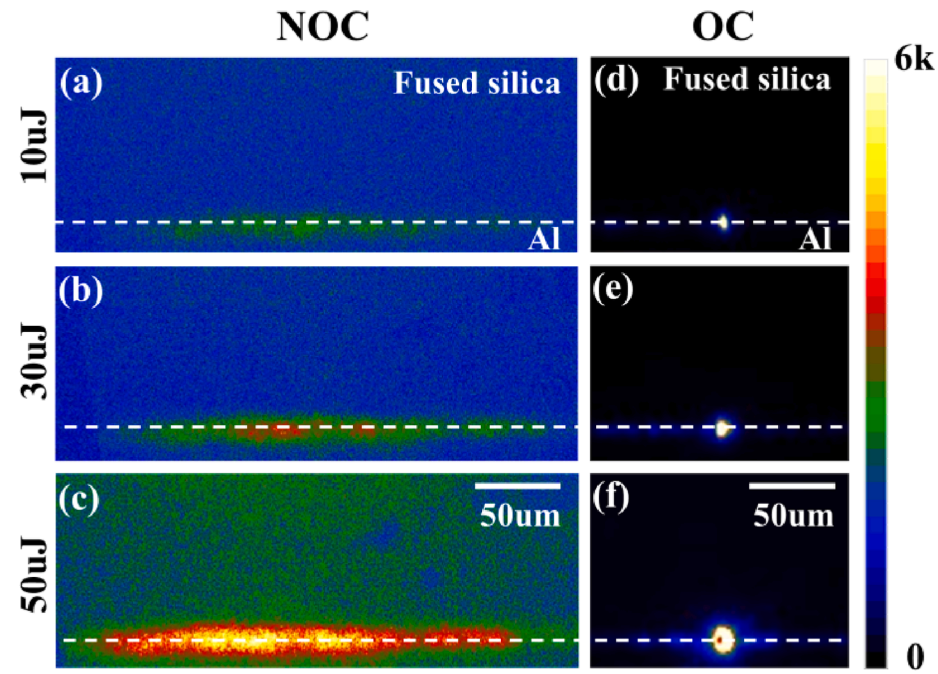

Fig. 2. Plasma behaviors at single pulse energy of 10, 30 and 50 μJ: (a–c) with the NOC interface and (d–f) with the OC interface. All the images were captured with 5 ns exposure time.

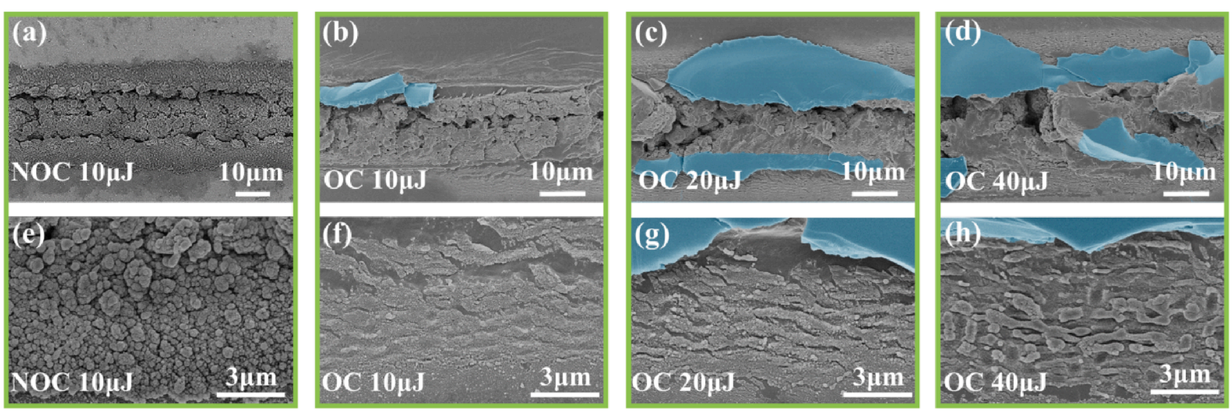

Fig. 3. SEM images of the aluminum surface with a scan speed of 70 μm/s. (a–d) Irradiation/weld seam morphology at the NOC and OC interfaces; (e–h) seam edge morphology at the NOC and OC interfaces. The blue parts are large chunks of fused silica adhering on the seam of the aluminum surface (the related elemental analysis is shown in Fig. S2 of the Supporting Information).

3. Results and discussion

3.1. Plasma images of the interaction area for non-optical contact (NOC) and optical contact (OC)

An ICCD was used to observe the plasma behaviors at NOC and OC interfaces, and the results are shown in Fig. 2. The value of the color scale represents the number of photons received per pixel on the sensor of ICCD. And the colors correspond to the relative intensity of the plasma signals. As shown in Fig. 2(a–c), for the NOC interface, the plasma generated by a single pulse diffused along the contact surface to the area outside the focus. The mirror-polished surfaces would reflect the plasma emitted light, so there is blue/green color in Fig. 2(a–c). There is visible luminescence below the dash line, which could be the plasma signals reflected by polished surfaces, and it would not affect the results and conclusions. Subsequently, the plasma cooled and formed nano-particles covering the irradiation region. The nanoparticles in Fig. 3(a and e) should be produced by laser ablation at NOC interface, which were similar to the surface nano-structures shown in Ref. [37]. As shown in Fig. 2(d–f), the plasma generated by a single pulse was strongly converged in the focal area for OC interface. The brighter plasma demonstrates much higher local temperature for OC than that of NOC.

As shown in Fig. 3, the microscopic structures at the welding seam with OC and NOC states were obviously different. As shown in Fig. 3(a and e), the edge and the center of the irradiation area with NOC were aggregated particles, while the edge with OC was a smaller area of corrugated microstructure. After the laser irradiation, the weld seam components were distributed in the focal area and on both sides. Consequently, it is challenging to form a stable connection for NOC because of the fast energy diffusion through the gap and fast temperature decrease, even with the pulse energy up to 50 μJ. As shown in Fig. 3 (b–d), more fused silica was coated on the weld seam with increased pulse energy at OC interface. It is speculated that the transient local pressure and heat at the focal area would lead to great temperature gradient, then the internal stress would be formed during cooling. The fused silica fractures and covers on the weld seams of the aluminum

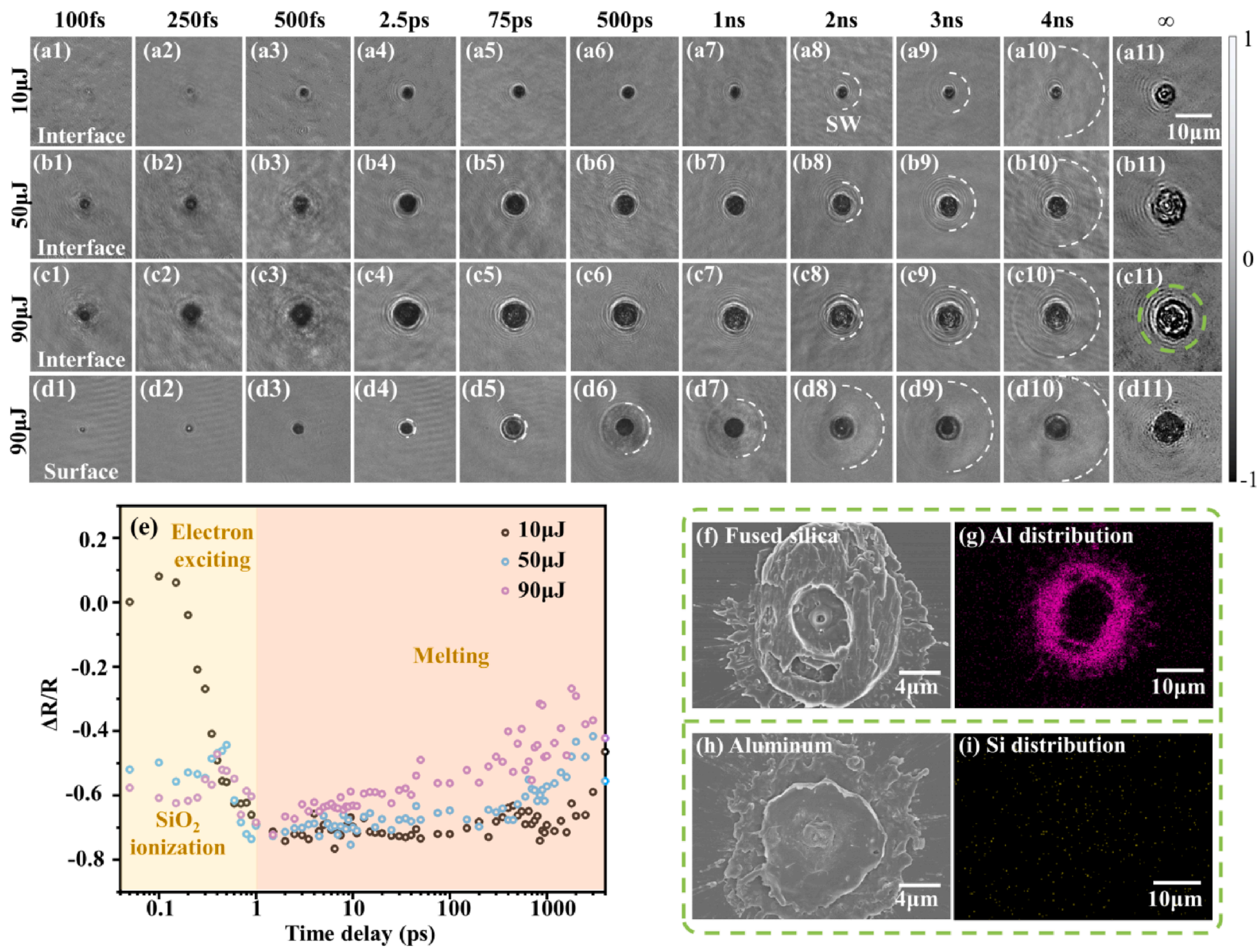

Fig. 4. Time-resolved reflectivity at the OC interface irradiated with different pulse energies: (a1–a11) 10 μJ, (b1–b11) 50 μJ, (c1–c11) 90 μJ, and (d1–d11) the surface of aluminum in the ambient air of 90 μJ with same experimental condition. (e) Normalized interface reflectivity change ΔR/R as a function of time, measured at the center of the irradiated regions with radius of 4 pixels. The error could be in the range of ± 0.03, and not shown in the figure. (f and g) and (h and i) are SEM images and EDS analysis with single pulse irradiated on the disassembled samples of fused silica and aluminum in (c11), respectively. Pink and yellow colors represent aluminum (Al) and silicon (Si), respectively.

sample during the shear strength test. The fused silica then fractures and covers the weld seams of the aluminum sample during the shear strength test. The aluminum melted and then condensed at the focal area, forming valid connections.

Meanwhile, the corrugated microstructures at the seam edges get a larger range and protrude more obviously with increased pulse energy, as shown in Fig. 3(f–h). The residual pressure and the heat after pulsed irradiation caused the structures. Previous studies have also reported corrugated structures with deeper groves, called "Keyholes" in picosecond laser welding [16]. Compared with picosecond laser, femtosecond laser does less damage at weld seams, with almost no micron cracks and pores.

3.2. *Time-resolved reflectivity with pump–probe microscopy at the OC interface*

The transient reflectance and morphologies obtained at the OC interface are shown in Fig. 4(a–c). The size of the interaction area increased slightly within the time range of 100 fs–∞. Diffraction rings appeared around the irradiated spot from 2.5 ps. And around 2 ns the shock waves expanded outward. The transient morphologies obtained at aluminum surface are shown in Fig. 4(d). Shock waves also appeared from 2.5 ps, but expanded rapidly in the ambient air. The shock wave expansion velocity at 3 ns could reach up to 5.00×10^3 m/s in air and 5.42×10^3 m/s at the OC interface. It is speculated that the shock waves observed at the OC interface could be generated inside the fused silica, which could be produced by the force released by the transient high pressure extruded lattice. The modified spot had a ring structure in the steady state, similar to the structure in liquid ablation in previous studies [31]. The common feature of the fused silica-to-aluminum welding and liquid ablation is that the metal surface was tightly covered with the transparent medium, making the melting material confined to a small area and causing non-free expansion at the irradiation spot. This means that the edge area of the modified region began to condense and solidify when the central area was still excited and expanded outward, resulting in a hole surrounded by a ring structure of aluminum element on the substrate surface of fused silica, as shown in Fig. 4(f and g). But there was no significant Si element on the corresponding spot at the aluminum surface, the main element of the circular structure could also be Al, as shown in Fig. 4(h and i). In the average center ΔR/R, the evolution process of reflectance was divided into two distinct periods before 1 ns, as shown in Fig. 4(e).

2 pictures would be taken during the experiment, one is before laser irradiation (the background), the second is at the specific time after laser irradiation. Pictures in Fig. 4 are the second ones minus the background. And we extracted the center reflectance R' of the second picture, R is the reflectance of same position in the background, ΔR/R=(R'-R)/R. When ΔR/R = 0, it means that the reflectance at specific time is consist with the background. When ΔR/R = 1, indicating the probe light is all reflected. When ΔR/R = −1, indicating the probe light is all absorbed.

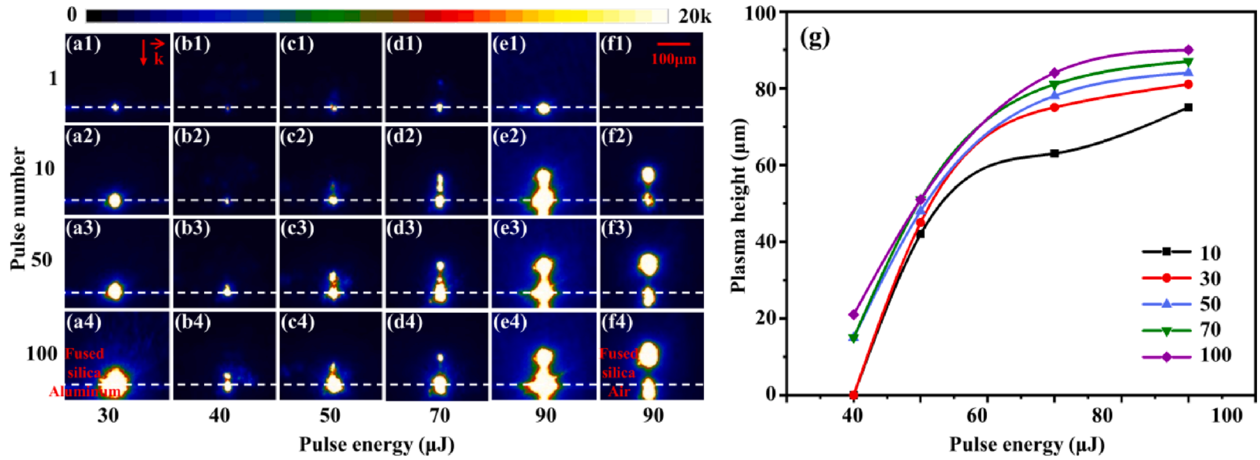

Fig. 5. Plasma behaviors recorded at the gate width of 5 ns after 1–100 pulses interact with the OC interface at different pulse energies: (a1–a4) 30 μJ, (b1–b4) 40 μJ, (c1–c4) 50 μJ, (d1–d4) 70 μJ, (e1–e4) 90 μJ, and (f1–f4) the substrate surface of fused silica without aluminum at pulse energy of 90 μJ under the same experimental condition. (g) The comparison of divided plasma heights with different pulse numbers (10–100 shots) and energies (40–90 μJ).

In the time scale of **0–1 ps**, the material would be excited when the laser pulse energy exceeds the ablation threshold. The ΔR/R evolution during the first several picoseconds at the OC interface is similar to that of aluminum surface [38,39]. The ionization in fused silica is not obvious and has minor effect on the ΔR/R evolution. And aluminum dominates the change in reflectivity. The temporary rise in reflectivity could be resulted to the interband transition caused by the excitation on aluminum surface [40,41]. The decreasing after 100 fs would be attributed to the electron temperature rise [42–44]: The decrease in electron temperature would result in a decrease in the electron collision time, causing variations in both the real and imaginary parts of the dielectric function. Consequently, the refractive index and extinction coefficient would exhibit changes that lead to a reduction in reflectivity. When the pulse energy increased to 50 and 90 μJ, the pulse would also

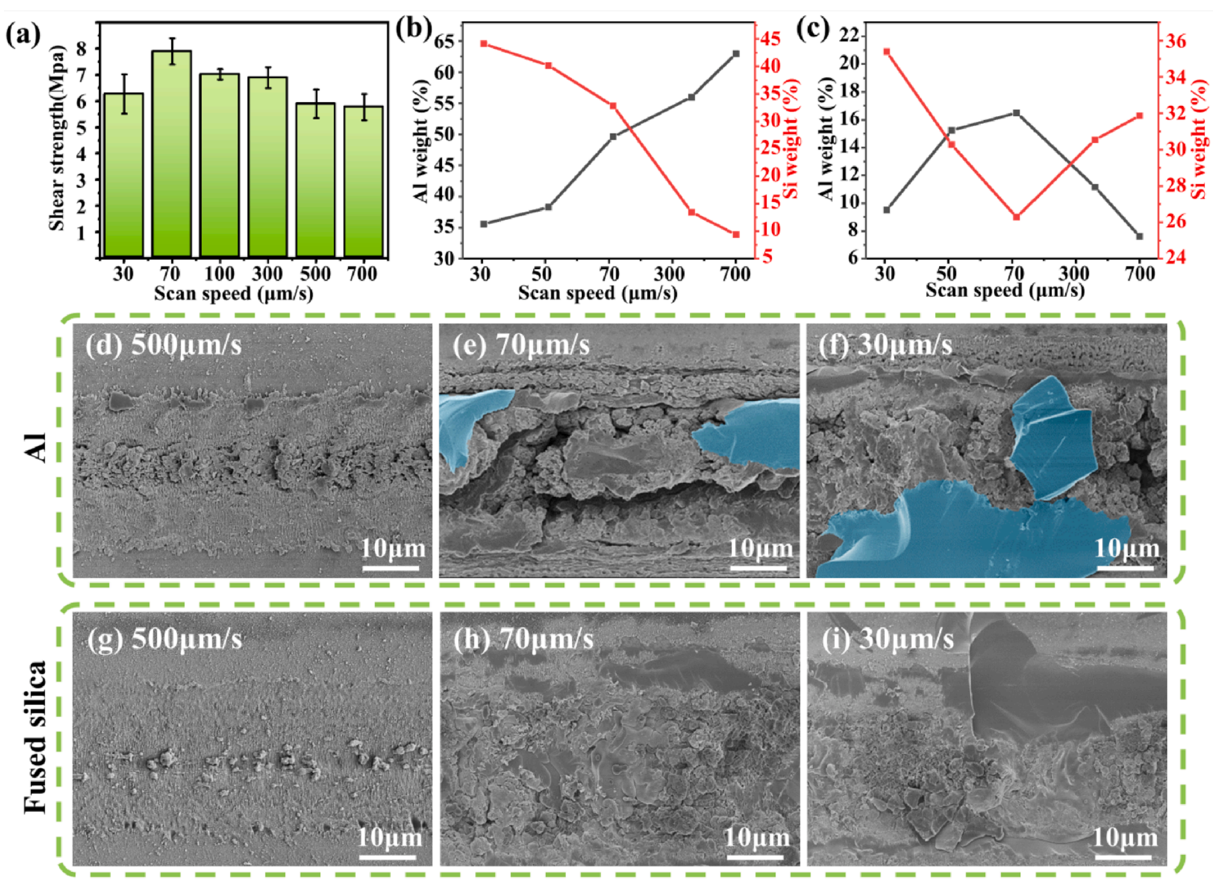

Fig. 6. For the same pulse energy of 20 μJ: (a) Relationship between the shear strength and welding speed, relationship between welding speed and the weld seam element (Al and Si) weight ratio on the (b)aluminum and (c)fused silica surface, (d–f) weld seam morphology on the aluminum surface for different speeds, and (g-i) weld seam morphology on the fused silica surface for different welding speeds. The blue parts represent large chunks of fused silica adhering on the seam of the aluminum surface.

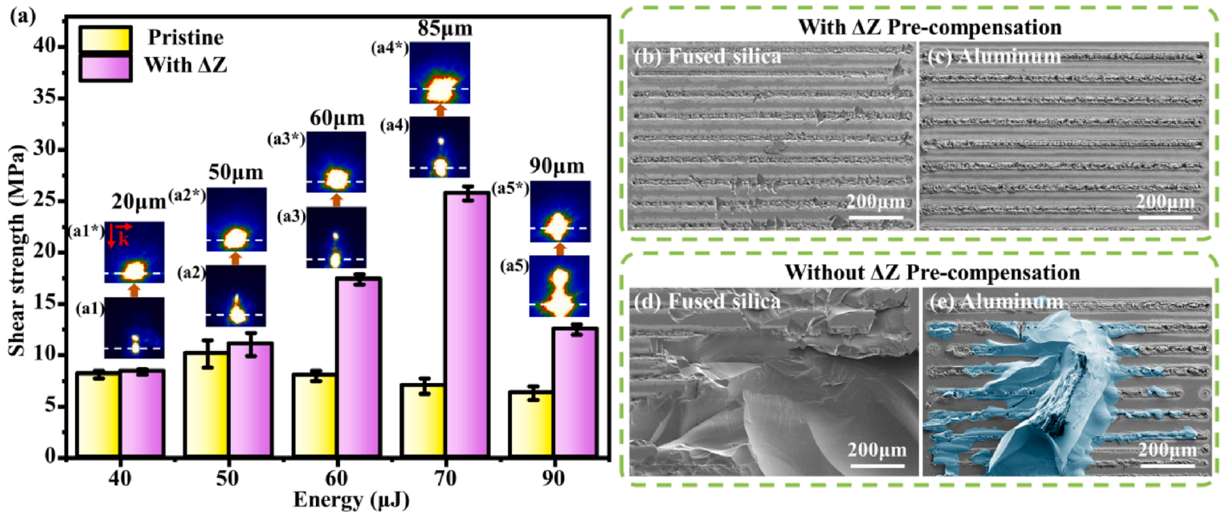

Fig. 7. (a) Shear strength with and without focus compensation (ΔZ) for different pulse energies: (a1*–a5*) and (a1–a5) are the plasma behaviors with and without ΔZ pre-compensation, respectively. (b–d) The surface of the disassembled samples with and without ΔZ pre-compensation on fused silica and aluminum with pulse energy of 50 μJ. The blue parts represent large chunks of fused silica cladding on the seam of the aluminum surface.

ionize the fused silica. Therefore, the increasing electron density leads to a strongly absorbing region [45] right above the contact interface inside the fused silica due to the self-focusing and non-linear absorption (caused by high pulse intensity). And the region would absorb the probe light, resulting in the decreased reflectance between 0 and 1 ps.

In the time scale of **1 ps–4 ns**, the energy absorbed by electrons is transferred rapidly to the lattice in a few picoseconds [46], leading to subsequent melt of aluminum [40]. As shown in Fig. 4(e), the reflectivity changed little for 10 μJ pulse energy after 1 ps. The reflectance rose slightly when the pulse energy increased to 50 μJ. Compared with a single pulse irradiating of aluminum surface in Fig. 4(d), the transient Newton ring morphology disappeared in Fig. 4(a-c) during 1–4 ns. It is speculated that a tightly covered fused silica could limit the motion, evaporation, and disintegration of molten aluminum at the focal point after laser irradiation. The reflectance rose significantly when the pulse energy increased to 90 μJ, which may be attributed to the different thermal conductivities of aluminum and fused silica, resulting in an obvious local non-equilibrium state at higher energy.

3.3. Plasma splitting effects in fused silica

Furthermore, plasma behaviors were obtained within 5 ns after irradiation of multiple pulses (1–100 shots) to investigate the welding process. Plasma splitting were observed inside the fused silica with increased energy. As shown in Fig. 5(a–e), the number of pulses was 1, 10, 50, and 100 respectively. The geometric focus of the pulses was set at the OC interface. Additionally, the plasma generated on the substrate surface of only fused silica was collected under the same experimental conditions, as shown in Fig. 5(f). The divided plasma heights (from the middle of the split plasma to the OC interface) with different pulse numbers and energies are shown in Fig. 5(g). As shown in Fig. 5(f1), forming luminous plasma with single pulse irradiation is challenging because of the high ablation threshold and multi-photon absorption inside fused silica. Therefore, the plasma was caused by the excitation of aluminum at the OC interface under single pulse irradiation, as shown in Fig. 5(a1–e1). The divided plasma appeared with the increase of pulse number. Former pulses could modify the interior fused silica and reduce the ablation threshold. Thus, bright plasma was observed by repeatedly exciting the modified region with more pulses. The plasma divided significantly at 40 μJ pulse energy, and the plasma intensity at the OC interface was significantly decreased compared with that of 30 μJ pulse energy. Therefore, the self-focusing effect would strongly weaken the energy deposition at the interface. Non-linear absorption would also occur at the time during the femtosecond laser passing through the fused silica. It would also affect the energy deposition efficiency at the OC interface.

Aluminum and fused silica could be both excited to form brighter plasma at the OC interface than that of only fused silica, as compared in Fig. 5(e) and (f). As shown in Fig. 5(g), the plasma height inside fused silica increased firstly and then flattened after 70 μJ with the same number of pulses, which shows that the plasma separation inside fused silica is limited due to the strong non-linear absorption. Furthermore, the intensity and height of the split plasma increased with pulse number, indicating that more energy was absorbed by fused silica and the internal modification and stress would be increased. It could be of little difference in split plasma (heights and sizes) between in Fig. 5(e1-e4) and Fig. 5(f1-f4). But the plasma at the OC interface in Fig. 5(e1-e4) are larger and brighter, which would be caused by additional plasma from the excited aluminum.

3.4. Welding morphology and shear strength

Firstly, low pulse energy of 20 μJ was used to explore the relationship between different welding speeds and shear strengths in the experiment. The shear strengths of aluminum and fused silica welded at different speeds (30–700 μm/s) are shown in Fig. 6(a). The shear strength increased and then decreased with the increase of the welding speed, and the maximum shear stress was 7.875 MPa at the welding speed of 70 μm/s. SEM and EDS were used to observe the surface morphology and analyze the elemental composition of samples after the shear test, as shown in Fig. 6(b–i). EDS analyze details are shown in Fig. S3 in the Supporting Information. From Fig. 6(b), it can be found that the Al/Si weight ratio continued to increase/decrease with the welding speed increased, and tended to be the same with the speed of 70 μm/s on the aluminum surface. From Fig. 6(c), Al/Si weight ratio reached the maximum/minimum on the fused silica surface with the speed of 70 μm/s, which corresponded to the maximum shear stress. Both results illustrated the higher mixing of Al and Si at the weld seam means high welding shear strength.

As shown in Fig. 6(d and g), fused silica is less affected by laser irradiation and obvious re-solidified aluminum particles were found in the middle area of the weld seam, which were melted and used to connect samples when the welding speed was 500 μm/s. As shown in Fig. 6(e and h) and Fig. S3, the silicon content increased obviously when

Table 1
Comparison of the shear strength of glass and metals/alloys welding by ultrafast laser.

Material	Samples' size(mm^3)	Maximum shear strength (MPa)	Laser wavelength; FWHM	PRF	Reference
Alumina-silicate glass-aluminum	5X10X0.820X20X20	2.34	800 nm, 160 fs	1 kHz	[47]
Fused silica-aluminum	–	1.43	800 nm, 10 ps	400 kHz	[48]
BK7 glass-Al6082	10X10X10 15X15 × 5	13	1030 nm, 5.9 ps	400 kHz	[49]
Fused silica-aluminum	**10X10X1**	**25.75**	**800 nm, 35 fs**	**1 kHz**	**This study**

the welding speed was about 70 μm/s, indicating that the fused silica was melted at this speed and was mixed with aluminum under the irradiation of the femtosecond laser. The interfusion of silicon and aluminum significantly increased the shear strength. Furthermore, more energy tended to deposit inside the fused silica, fractured easily with shear force and adhered on the aluminum surface when the welding speed went down to 30 μm/s, as shown in Fig. 6(f and i).

For laser welding at higher energy, a pre-compensated distance ΔZ for the focus position were applied to reduce the energy absorption inside the fused silica, which means move the focus position down with the distance corresponding to the plasma height in Fig. 5(g) at 100 pulses irradiation. The shear strength with and without ΔZ for different laser energies is shown in Fig. 7(a). The shear strength could be significantly increased by focal point pre-compensation at pulse energies of 50 μJ and above. The maximum shear strength of the welded sample could reach 25.75 MPa (at pulse energy of 70 μJ and ΔZ of 85 μm), more than twice as much as before pre-compensation. The focus pre-compensation changed the plasma behaviors at the OC interface, resulting in plasma that no longer splitted and luminous volume increased, as shown in Fig. 7(a1*–a5*). After the shear test, the state of fused silica with and without ΔZ pre-compensation was significantly different. The fused silica could be completely disassembled with ΔZ, and it was easy to break down during the test without ΔZ, as shown in Fig. S4. Compared with the reported studies on laser welding of similar objects (Table 1), the shear strength was quite superior. The shear strength decreased at 90 μJ energy because the modification was so severe inside fused silica that could not be avoid by pre-compensation, and the increase of internal stress would reduce the shear resistance of the welded sample.

The comparison of the disassembled morphology at pulse energy of 50 μJ with and without the pre-compensation of the focal position is shown in Fig. 7(b–e). Without focus pre-compensation, the fused silica was easily flaked off in large pieces at the center of the square welding area. The significantly increased shear strength and SEM morphology differences indicate that the focus pre-compensation could increase the pulse energy deposition at the OC interface and avoid the modification to the greatest extent of the fused silica interior.

4. Conclusion

The transient interaction between femtosecond laser and the OC interface of heterogeneous materials was investigated using aluminum and fused silica. Time-resolved reflectivity showed the different dominant factors, which are electron heating of aluminum at lower energy and ionization inside fused silica at higher pulse energy before 1 ps. In the time scale of 1 ps–4 ns, melting begins and the reflectivity rises again at high pulse energy. The ring structure appeared in steady state due to the non-free expansion of the melting material at the OC interface. The plasma images with multiple pulses and different energies at the OC interface shows that the self-focusing effects formed the splitting plasma inside the fused silica. Once the plasma is split, the beam propagation would be obstructed and the energy deposition at the OC interface dropped dramatically. According to the captured plasma heights after 100 pulses irradiation, the focus position during the welding process was pre-compensated. It was found that the maximum shear strength was 25.75 MPa at the pulse energy of 70 μJ with a pre-compensation distance of 85 μm. With higher pulse energy, the inner stress was hard to be avoided because of strong non-linear absorption. This work demonstrates that appropriate pulse energy and focus pre-compensation can increase the energy deposition efficiency at the welding interface and reduce the internal modification inside fused silica, which could improve the shear strength of the welded samples.

CRediT authorship contribution statement

Jie Zhan: Data curation, Investigation, Writing - original draft. **Yuhang Gao:** Formal analysis; Validation. **Jiaxin Sun:** Methodology. **Weihua Zhu:** Conceptualization. **Sumei Wang:** Writing - review & editing, Funding aquisition. **Lan Jiang:** . **Xin Li:** Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the National Key R&D Program of China (2022YFB4600400)

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apsusc.2023.158327.

References

[1] S. Kim, J. Park, S. So, S. Ahn, J. Choi, C. Koo, Y.-H. Joung, Characteristics of an implantable blood pressure sensor packaged by ultrafast laser microwelding, Sensors 19 (2019) 1801.
[2] C. Yang, C. Zhao, L. Wold, K.R. Kaufman, Biocompatibility of a physiological pressure sensor, Biosens. Bioelectron.. 19 (2003) 51–58.
[3] Y. Temiz, R.D. Lovchik, G.V. Kaigala, E. Delamarche, Lab-on-a-chip devices: How to close and plug the lab? Microelectron. Eng. 132 (2015) 156–175.
[4] K. Tamrin, Y. Nukman, S. Zakariyah, Laser lap joining of dissimilar materials: a review of factors affecting joint strength, Mater. Manuf. Process. 28 (2013) 857–871.
[5] H. Kato, S.-G. Yu, A.D. Setyawan, K. Yubuta, T. Wada, Surface-activated supercooled liquid brazing, Scr. Mater. 68 (2013) 699–702.
[6] M. Wild, A. Gillner, R. Poprawe, Locally selective bonding of silicon and glass with laser, SNA 93 (2001) 63–69.
[7] P. Kah, R. Suoranta, J. Martikainen, C. Magnus, Techniques for joining dissimilar materials: metals and polymers, Rev. Adv. Mater. Sci. 36 (2014).
[8] P. Zhang, Z. Jia, Z. Yu, H. Shi, S. Li, D. Wu, H. Yan, X. Ye, J. Chen, F. Wang, A review on the effect of laser pulse shaping on the microstructure and hot cracking behavior in the welding of alloys, Opt. Laser Technol. 140 (2021), 107094.
[9] J. Huo, B. Zhang, C. Li, M. Luo, T. Chen, L. Guo, A. Luo, N. Zhao, J. Li, Q. Zhang, The mechanism of the welding between silica glass and 304 stainless steel using nanosecond fibre laser, Sci. Technol. Weld Jo I (2023) 1–8.
[10] J. Yang, J. Oliveira, Y. Li, C. Tan, C. Gao, Y. Zhao, Z. Yu, Laser techniques for dissimilar joining of aluminum alloys to steels: A critical review, J. Mater. Process. Technol. 301 (2022), 117443.
[11] C. Li, J. Tan, M. Luo, W. Chen, Y. Huang, J. Gu, N. Zhao, J. Li, H. Yang, Q. Zhang, High shear strength welding of soda lime glass to stainless steel using an infrared

nanosecond fiber laser assisted by surface tension, Opt. Laser Technol. 161 (2023), 107329.
[12] S. Yan, Z. Li, L. Song, Y. Zhang, S. Wei, Research and development status of laser micro-welding of aluminum-copper dissimilar metals: A review, Opt. Laser Technol. 161 (2023), 107312.
[13] T.O. Lipateva, A.G. Okhrimchuk, A.S. Lipatiev, I.A. Karateev, S.S. Fedotov, S. V. Lotarev, A.V. Shestakov, V.N. Sigaev, Robust and adhesive-free joint of Nd: YAG crystals by femtosecond laser-assisted welding, Opt. Laser Technol. 146 (2022), 107594.
[14] E. Penilla, L. Devia-Cruz, A. Wieg, P. Martinez-Torres, N. Cuando-Espitia, P. Sellappan, Y. Kodera, G. Aguilar, J. Garay, Ultrafast laser welding of ceramics, Science 365 (2019) 803–808.
[15] J. Zhang, S. Xu, Y. Dong, C. Zhang, J. Li, L. Guo, Q. Zhang, Microwelding of glass to silicon by green ultrafast laser pulses, Opt. Laser Technol. 120 (2019), 105720.
[16] O.P. Ciuca, R.M. Carter, P.B. Prangnell, D.P. Hand, Characterisation of weld zone reactions in dissimilar glass-to-aluminium pulsed picosecond laser welds, Mater. Charact. 120 (2016) 53–62.
[17] A. de Pablos-Martín, M. Lorenz, M. Grundmann, T. Höche, Laser welding of fused silica glass with sapphire using a non-stoichiometric, fresnoitic Ba2TiSi2O8· 3 SiO2 thin film as an absorber, Opt. Laser Technol. 92 (2017) 85–94.
[18] G. Zhang, J. Bai, W. Zhao, K. Zhou, G. Cheng, Interface modification based ultrashort laser microwelding between SiC and fused silica, Opt. Express 25 (2017) 1702–1709.
[19] M. Chambonneau, Q. Li, V.Y. Fedorov, M. Blothe, K. Schaarschmidt, M. Lorenz, S. Tzortzakis, S. Nolte, Taming ultrafast laser filaments for optimized semiconductor–metal welding, Laser Photonics Rev. 15 (2021) 2000433.
[20] K. Cvecek, S. Dehmel, I. Miyamoto, M. Schmidt, A review on glass welding by ultra-short laser pulses, IJEM 1 (2019), 042001.
[21] S. Richter, F. Zimmermann, A. Tünnermann, S.J.O. Nolte, L. Technology, Laser welding of glasses at high repetition rates–Fundamentals and prospects, Opt. Laser Technol. 83 (2016) 59–66.
[22] A. Couairon, A. Mysyrowicz, Femtosecond filamentation in transparent media, Phys. Rep. 441 (2007) 47–189.
[23] I. Miyamoto, K. Cvecek, M. Schmidt, Evaluation of nonlinear absorptivity in internal modification of bulk glass by ultrashort laser pulses, Opt. Express 19 (2011) 10714–10727.
[24] J.R. Gulley, S.W. Winkler, W.M. Dennis, C.M. Liebig, R. Stoian, Interaction of ultrashort-laser pulses with induced undercritical plasmas in fused silica, Phys. Rev. A 85 (2012), 013808.
[25] F. Wang, C. Pan, J. Sun, Q. Wang, Y. Lu, L. Jiang, Direct observation of structure-assisted filament splitting during ultrafast multiple-pulse laser ablation, Opt. Express 27 (2019) 10050–10057.
[26] C. Pan, L. Jiang, J. Sun, Q. Wang, F. Wang, K. Wang, Y. Lu, Y. Wang, L. Qu, T. Cui, Ultrafast optical response and ablation mechanisms of molybdenum disulfide under intense femtosecond laser irradiation, Light Sci. Appl. 9 (2020) 80.
[27] Y. Lian, Y. Hua, J. Sun, Q. Wang, Z. Chen, F. Wang, K. Zhang, G. Lin, Z. Yang, Q. Zhang, J. Lan, Martensitic transformation in temporally shaped femtosecond laser shock peening 304 steel, Appl. Surf. Sci. 567 (2021), 150855.
[28] J. Song, J. Sun, Y. Lian, W. Tao, Y. Xie, Y. Yang, High-performance antireflection nanostructure arrays on aluminum-doped zinc oxide film fabricated with femtosecond laser near-field processing, Appl. Surf. Sci. 156353 (2023).
[29] Z. Ning, Y. Lian, L. Jiang, J. Sun, S. Wu, F. Wang, Femtosecond laser-induced anisotropic structure and nonlinear optical response of yttria-stabilized zirconia single crystals with different planes, ACS Appl. Mater. Inter. 14 (2022) 39591–39600.
[30] Y. Furukawa, M. Hashida, S. Kojima, S. Inoue, S. Sakabe, Optical properties of titanium induced by below-ablation-threshold irradiation, Appl. Surf. Sci. 515 (2020), 146047.
[31] A. Kanitz, D.J. Förster, J.S. Hoppius, R. Weber, A. Ostendorf, E.L. Gurevich, Pump-probe microscopy of femtosecond laser ablation in air and liquids, Appl. Surf. Sci. 475 (2019) 204–210.
[32] S.I. Kudryashov, A.A. Samokhvalov, Y.D. Golubev, D.S. Ivanov, M.E. Garcia, V. P. Veiko, B. Rethfeld, V.Y. Mikhailovskii, Dynamic all-optical control in ultrashort double-pulse laser ablation, Appl. Surf. Sci. 537 (2021), 147940.
[33] Y. Lian, L. Jiang, J. Sun, J. Zhou, Y. Zhou, Ultrafast Quasi-Three-Dimensional Imaging, IJEM, (2023).
[34] P. Sopeña, A. Wang, A. Mouskeftaras, D. Grojo, Transmission laser welding of similar and dissimilar semiconductor materials, Laser Photonics Rev. 16 (2022) 2200208.
[35] H. Chen, J. Duan, Z. Yang, W. Xiong, L. Deng, Picosecond laser seal welding of glasses with a large gap, Opt. Express 27 (2019) 30297–30307.
[36] H. Tan, J.a. Duan,, Welding of glasses in optical and partial-optical contact via focal position adjustment of femtosecond-laser pulses at moderately high repetition rate, Appl. Phys. A 123 (2017) 1–9.
[37] H. Yang, H. He, E. Zhao, J. Qian, J. Hao, W. Tang, H. Zhu, Laser fabrication of nanostructures on a pure aluminum surface in air, J. Russ. Laser Res. 34 (2013) 362–368.
[38] M. Spellauge, J. Winter, S. Rapp, C. McDonnell, F. Sotier, M. Schmidt, H.P. Huber, Influence of stress confinement, particle shielding and re-deposition on the ultrashort pulse laser ablation of metals revealed by ultrafast time-resolved experiments, Appl. Surf. Sci. 545 (2021), 148930.
[39] M. Domke, S. Rapp, M. Schmidt, H.P. Huber, Ultrafast pump-probe microscopy with high temporal dynamic range, Opt. Express 20 (2012) 10330–10338.
[40] M. Kandyla, T. Shih, E. Mazur, Femtosecond dynamics of the laser-induced solid-to-liquid phase transition in aluminum, Phys. Rev. B 75 (2007), 214107.
[41] D. Fisher, M. Fraenkel, Z. Henis, E. Moshe, S. Eliezer, Interband and intraband (Drude) contributions to femtosecond laser absorption in aluminum, Phys. Rev. E 65 (2001), 016409.
[42] G. Lin, P. Ji, M. Wang, Y. Meng, Numerical insight into heat transfer in surface melting and ablation subject to femtosecond laser processing aluminum, Int. Commun. Heat Mass 142 (2023), 106649.
[43] S. Nagel, S. Schnatterly, Frequency dependence of the Drude relaxation time in metal films, Phys. Rev. B 9 (1974) 1299.
[44] L. Jiang, H.-L. Tsai, Energy transport and nanostructuring of dielectrics by femtosecond laser pulse trains, Int Commun Heat Mass, (2006).
[45] M. García-Lechuga, L. Haahr-Lillevang, J. Siegel, P. Balling, S. Guizard, Solis, Simultaneous time-space resolved reflectivity and interferometric measurements of dielectrics excited with femtosecond laser pulses, Phys. Rev. B 95 (2017), 214114.
[46] C. Fourment, F. Deneuville, B. Chimier, D. Descamps, F. Dorchies, S. Hulin, S. Petit, O. Peyrusse, J. Santos, An interferometric diagnostic for the experimental study of dynamics of solids exposed to intense and ultrashort radiation, in: Damage to VUV, EUV, and X-ray Optics IV; and EUV and X-ray Optics: Synergy Between Laboratory and Space III, SPIE, 2013, pp. 129–139.
[47] G. Zhang, G. Cheng, Direct welding of glass and metal by 1 kHz femtosecond laser pulses, Appl. Opt. 54 (2015) 8957–8961.
[48] R.M. Carter, J. Chen, J.D. Shephard, R.R. Thomson, D.P. Hand, Picosecond laser welding of similar and dissimilar materials, Appl. Opt. 53 (2014) 4233–4238.
[49] R.M. Carter, M. Troughton, J. Chen, I. Elder, R.R. Thomson, M.D. Esser, R.A. Lamb, D.P. Hand, Towards industrial ultrafast laser microwelding: SiO 2 and BK7 to aluminum alloy, Appl. Opt. 56 (2017) 4873–4881.